MICROSOFT SECURITY OPERATIONS & DEFENDER SENTINEL PROFESSIONAL

Beginner-Friendly Practical Training Program

60 Hours (3 months) • *Microsoft Sentinel* • *Defender XDR* • *Defender for Endpoint* • *Entra ID* • *KQL* • *Incident Response*

Recommended Duration	60 Hours
Level	Beginner → Job-Ready Microsoft Security Operations Foundation
Primary Environment	Microsoft Defender portal + Microsoft Sentinel + Microsoft 365 E5 trial lab
Long-Term Practice	Microsoft Learn labs + local Windows/Linux + Wazuh/open-source alternatives

Course Positioning

This is an independent practical training program. It is not an official Microsoft course and does not reproduce Microsoft courseware. The course is designed around the skills a beginner needs to operate Microsoft security tools in a SOC, with emphasis on Sentinel, Defender XDR, Defender for Endpoint, Microsoft Entra ID, KQL, threat hunting and incident response.

Microsoft's current Security Operations Analyst learning material covers Microsoft Sentinel and Microsoft Defender XDR, while the current certification study guidance also references Microsoft security, compliance and identity solutions, Microsoft 365, Azure, AI agents/Copilots, and Windows/Linux/mobile environments. This course converts those broad capabilities into a beginner-friendly, hands-on lab sequence. □cite□turn0search20□turn0search15□

Recommended Total Teaching Hours

Recommended instructor-led duration: 60 hours. For beginners, this gives enough time to configure the trial environment, repeat KQL investigations, practice Defender incidents, perform Sentinel investigations and complete an end-to-end SOC simulation.

Area	Hours	Theory / Lab
Microsoft Security & SOC Foundations	6	3 + 3
Microsoft Cloud & Identity Foundations	6	3 + 3
Defender Portal & Defender XDR	8	4 + 4
Defender for Endpoint	8	3 + 5
Defender for Office 365	4	2 + 2
Microsoft Entra ID Security	4	2 + 2
Microsoft Sentinel	12	5 + 7
KQL & Advanced Hunting	5	2 + 3
Threat Intelligence / Hunting / IR	4	2 + 2
Automation & AI-Assisted SOC	1	0.5 + 0.5
Final SOC Simulation	2	0.5 + 1.5

TOTAL: 60 HOURS

Current Microsoft Trial Strategy

Microsoft Sentinel currently provides a 31-day free trial when enabled on an Azure Monitor Log Analytics workspace; Microsoft documents the first 10 GB/day of Analytics-log ingestion as free during that trial. Additional capabilities can have separate charges. Sentinel is also generally available in the Microsoft Defender portal. □cite□turn0search0□turn0search17□

Microsoft provides a dedicated Defender XDR trial-lab setup based on a Microsoft 365 E5 trial tenant. Microsoft also documents Defender for Endpoint deployment and investigation labs using trial licenses. □cite□turn0search1□turn0search2□

Defender for Office 365 currently has a separate 90-day Plan 2 trial according to Microsoft documentation, but the course does not depend on that longer trial; the core learning path is designed around the one-month training window. □cite□turn0search7□

Trial availability, licensing, quotas and feature access can change by tenant, region and date. The instructor should verify the current trial terms before starting a batch and disable/remove chargeable Azure resources when labs are complete.

Recommended Lab Environment

- Dedicated Microsoft 365 trial/lab tenant rather than a production tenant.
- Microsoft 365 E5 trial where eligible for Defender XDR components.
- Microsoft Defender portal as the primary security console.
- Microsoft Sentinel connected to a Log Analytics workspace and/or Defender portal.
- One Windows 10/11 evaluation or licensed VM for Defender for Endpoint exercises where supported.
- Optional Ubuntu/Linux VM for cross-platform investigation.
- Azure subscription for Sentinel labs; monitor usage and trial limits.
- Optional Wazuh/local logs as a free alternative for students after Microsoft trials expire.
- Safe, synthetic identities, alerts and test data.

Course Roadmap

Module	Module Name	Core Focus
01	Microsoft Security Operations Foundations	SOC roles, Microsoft security ecosystem, threats and controls
02	Microsoft Cloud & Security Architecture Basics	Azure, Microsoft 365, Defender portal and security data flow
03	Microsoft Entra ID for Security Analysts	Users, groups, authentication, sign-in logs and risky identities
04	Defender Portal & Unified Security Operations	Incidents, alerts, assets, users and investigation workflow
05	Microsoft Defender XDR Fundamentals	Cross-domain detection and incident correlation
06	Defender for Endpoint Setup & Telemetry	Onboarding, endpoint signals, device inventory
07	Endpoint Alert Investigation & Response	Process, file, device, user and response actions
08	Defender for Office 365 Security Operations	Email threats, URLs, attachments and investigation
09	Microsoft Sentinel Fundamentals	Workspace, connectors, analytics, incidents and workbooks
10	KQL for Security Analysts	Filtering, parsing, aggregation, joins and time analysis
11	Sentinel Analytics & Detection Rules	Scheduled/near-real-time rules, tuning and incidents
12	Sentinel Incident Investigation	Entities, evidence, bookmarks, timelines and case workflow
13	Advanced Hunting Across Microsoft Security	Cross-domain KQL and Defender telemetry
14	Threat Intelligence & Microsoft Security Data	Indicators, enrichment and contextual investigation
15	Threat Hunting with Microsoft Tools	Hypothesis-driven hunts and behavioral queries
16	MITRE ATT&CK Mapping & Detection Thinking	Tactics, techniques and coverage
17	Incident Response & Automated	Containment, remediation,

	Response	automation and playbooks
18	Security Automation & AI Assistance	Logic Apps concepts, Python/local AI and human approval
19	End-to-End Microsoft SOC Challenge	Multi-stage incident investigation
20	Final Capstone: Microsoft Security Operations Analyst	Build, investigate, respond and report

Detailed Module Content

Module 01: Microsoft Security Operations Foundations — 3 Hours

Theory Concepts

- SOC purpose and L1/L2/L3 responsibilities
- Threats, vulnerabilities, incidents and alerts
- Microsoft security ecosystem overview
- SIEM, XDR, EDR and identity-security concepts
- Incident lifecycle and analyst documentation

Hands-On Lab Objective

Understand the Microsoft security operations workflow before touching the tools.

Practical Lab Activities

1. Classify sample alerts.
2. Build a simple SOC escalation matrix.
3. Identify which Microsoft tool would provide each evidence type.
4. Create a sample incident ticket.

Microsoft / Lab Tools

Microsoft Defender portal, Microsoft Learn, LibreOffice

Expected Learning Outcome

Students understand the role of each major Microsoft security service.

Module 02: Microsoft Cloud & Security Architecture Basics — 3 Hours

Theory Concepts

- Azure subscription/resource concepts
- Microsoft 365 tenant concepts
- Identity, endpoint, email and cloud signals
- Defender portal architecture
- Security data flow into Sentinel

Hands-On Lab Objective

Map security telemetry from an endpoint, identity and cloud service into the Microsoft security ecosystem.

Practical Lab Activities

5. Create a simple Microsoft security architecture diagram.
6. Identify telemetry sources.
7. Explain Sentinel vs Defender XDR roles.
8. Identify where an L1 analyst starts an investigation.

Microsoft / Lab Tools

Microsoft Defender portal, Azure portal, diagrams.net

Expected Learning Outcome

Students can explain the Microsoft security architecture at a beginner level.

Module 03: Microsoft Entra ID for Security Analysts — 3 Hours

Theory Concepts

- Users, groups and roles
- Authentication and MFA concepts
- Sign-in logs
- Conditional Access concepts
- Identity risk and risky users
- Least privilege

Hands-On Lab Objective

Investigate identity-related activity in a safe Microsoft 365 lab tenant.

Practical Lab Activities

9. Create test users.
10. Generate successful/failed sign-ins where possible.
11. Review sign-in logs.
12. Identify unusual authentication patterns.
13. Review available risk signals.
14. Document a suspicious identity investigation.

Microsoft / Lab Tools

Microsoft Entra admin center, Microsoft Defender portal

Expected Learning Outcome

Students can investigate common identity-security events.

Module 04: Defender Portal & Unified Security Operations — 3 Hours

Theory Concepts

- Microsoft Defender portal navigation
- Incidents vs alerts
- Entities and evidence
- Incident queues
- Investigation workflow
- Response actions

Hands-On Lab Objective

Learn the primary analyst workflow in the Microsoft Defender portal.

Practical Lab Activities

15. Explore incident queue.
16. Open an incident.
17. Review alert evidence.
18. Pivot to user/device/email evidence.
19. Record findings.
20. Practice incident closure notes.

Microsoft / Lab Tools

Microsoft Defender portal, Microsoft 365 E5 trial

Expected Learning Outcome

Students can navigate and work a Defender incident.

Module 05: Microsoft Defender XDR Fundamentals — 4 Hours

Theory Concepts

- XDR vs SIEM
- Cross-domain correlation
- Identity, endpoint, email and application signals
- Incident correlation
- Advanced hunting concept
- Response capabilities

Hands-On Lab Objective

Investigate a simulated multi-signal incident in Defender XDR.

Practical Lab Activities

21. Review an incident containing multiple alerts.
22. Correlate user/device/email evidence.
23. Identify the probable attack path.
24. Determine affected entities.
25. Document recommended response.

Microsoft / Lab Tools

Microsoft Defender XDR, Microsoft 365 E5 trial

Expected Learning Outcome

Students understand how XDR reduces investigation fragmentation.

Module 06: Defender for Endpoint Setup & Telemetry — 4 Hours

Theory Concepts

- Endpoint protection concepts
- Device onboarding
- Security policies
- Endpoint telemetry
- Device inventory
- Attack surface reduction concepts

Hands-On Lab Objective

Onboard a supported lab endpoint and explore endpoint telemetry.

Practical Lab Activities

26. Prepare a Windows lab device.
27. Onboard where trial/licensing supports it.
28. Review device inventory.
29. Explore device timeline.
30. Review security configuration.
31. Generate safe test telemetry.

Microsoft / Lab Tools

Microsoft Defender for Endpoint, Windows VM, Microsoft 365 E5 trial

Expected Learning Outcome

Students can explain endpoint onboarding and telemetry.

Module 07: Endpoint Alert Investigation & Response — 4 Hours

Theory Concepts

- Process trees
- File and hash evidence
- Network connections
- Persistence concepts
- Device isolation concepts
- Automated investigation/remediation concepts

Hands-On Lab Objective

Investigate a controlled endpoint alert and determine appropriate response.

Practical Lab Activities

32. Review process/file evidence.
33. Pivot to device timeline.
34. Check user context.
35. Review network connections.
36. Determine whether the alert is true positive or benign.
37. Practice response decision-making.

Microsoft / Lab Tools

Defender for Endpoint, Defender XDR

Expected Learning Outcome

Students can investigate endpoint alerts and recommend response actions.

Module 08: Defender for Office 365 Security Operations — 4 Hours

Theory Concepts

- Email-borne threats
- Phishing
- Malicious links
- Attachments
- Sender authentication concepts
- Email investigation
- Threat Explorer concepts

Hands-On Lab Objective

Perform a safe simulated phishing investigation.

Practical Lab Activities

38. Review a sample message.
39. Inspect sender/recipient information.
40. Analyze URLs and attachment metadata.

41. Search related messages where trial features permit.
42. Identify affected users.
43. Create a response recommendation.

Microsoft / Lab Tools

Microsoft Defender for Office 365, Microsoft 365 E5 trial, CyberChef

Expected Learning Outcome

Students can triage common email-security incidents.

Module 09: Microsoft Sentinel Fundamentals — 4 Hours

Theory Concepts

- Log Analytics workspace
- Sentinel architecture
- Data connectors
- Tables and logs
- Analytics rules
- Incidents
- Workbooks
- Content hub concepts

Hands-On Lab Objective

Configure Sentinel and onboard a small amount of lab data within the trial allowance.

Practical Lab Activities

44. Create/use a Log Analytics workspace.
45. Enable Sentinel.
46. Explore tables.
47. Connect supported data sources.
48. Review ingested events.
49. Create a basic workbook panel.

Microsoft / Lab Tools

Microsoft Sentinel, Azure Monitor / Log Analytics

Expected Learning Outcome

Students can configure and navigate a beginner Sentinel environment.

Module 10: KQL for Security Analysts — 4 Hours

Theory Concepts

- KQL syntax
- where
- project
- extend
- summarize
- sort
- parse
- time filters
- joins

- let concepts

Hands-On Lab Objective

Write KQL queries for common SOC investigations.

Practical Lab Activities

50. Find failed logins.
51. Find top source IPs.
52. Summarize events by user.
53. Extract fields from text.
54. Compare time windows.
55. Join related datasets where available.
56. Save useful hunting queries.

Microsoft / Lab Tools

Microsoft Sentinel, Microsoft Defender Advanced Hunting, KQL

Expected Learning Outcome

Students can write practical beginner-to-intermediate KQL.

Module 11: Sentinel Analytics & Detection Rules — 3 Hours

Theory Concepts

- Analytics-rule purpose
- Scheduled rules
- Near-real-time concepts
- Thresholds
- Entity mapping
- Incident creation
- Tuning

Hands-On Lab Objective

Create a simple detection and tune it against controlled events.

Practical Lab Activities

57. Create an analytics rule.
58. Generate or ingest test events.
59. Review generated incidents.
60. Tune noisy conditions.
61. Map entities.
62. Document detection logic and limitations.

Microsoft / Lab Tools

Microsoft Sentinel, KQL

Expected Learning Outcome

Students can build and tune a basic Sentinel detection.

Module 12: Sentinel Incident Investigation — 3 Hours

Theory Concepts

- Incident queue
- Entities
- Investigation graph
- Timeline/evidence
- Bookmarks
- Comments and collaboration
- Incident status

Hands-On Lab Objective

Work a Sentinel incident from alert to documented conclusion.

Practical Lab Activities

63. Open incident.
64. Review entities.
65. Run supporting KQL.
66. Bookmark evidence.
67. Build a timeline.
68. Classify incident.
69. Document closure or escalation.

Microsoft / Lab Tools

Microsoft Sentinel, KQL

Expected Learning Outcome

Students can conduct a complete Sentinel incident investigation.

Module 13: Advanced Hunting Across Microsoft Security — 4 Hours

Theory Concepts

- Advanced Hunting workspace
- Device, identity, email and cloud telemetry
- Cross-table investigation
- Behavioral searches
- Hunting queries
- Query optimization

Hands-On Lab Objective

Use KQL to correlate signals across Microsoft security data.

Practical Lab Activities

70. Hunt suspicious processes.
71. Hunt unusual sign-ins.
72. Search email/URL indicators.
73. Correlate device and identity activity.
74. Build a reusable hunting query.
75. Document findings.

Microsoft / Lab Tools

Microsoft Defender Advanced Hunting, KQL

Expected Learning Outcome

Students can perform cross-domain threat hunting.

Module 14: Threat Intelligence & Microsoft Security Data — 2 Hours

Theory Concepts

- IOC types
- Indicator confidence
- Reputation and enrichment
- Threat context
- Microsoft threat intelligence concepts
- IOC limitations

Hands-On Lab Objective

Enrich a set of synthetic indicators and add context to an investigation.

Practical Lab Activities

76. Extract IP/domain/hash indicators.
77. Review permitted reputation information.
78. Compare context sources.
79. Assign confidence.
80. Add useful context to an incident report.

Microsoft / Lab Tools

Microsoft Defender, Microsoft Sentinel, VirusTotal where appropriate

Expected Learning Outcome

Students can enrich and contextualize indicators safely.

Module 15: Threat Hunting with Microsoft Tools — 3 Hours

Theory Concepts

- Alert-driven vs hypothesis-driven investigation
- Baselines
- Rare events
- Suspicious authentication
- Persistence
- Discovery and lateral movement

Hands-On Lab Objective

Perform a hypothesis-driven hunt using KQL.

Practical Lab Activities

81. Write a hunting hypothesis.
82. Search authentication anomalies.
83. Search suspicious process behavior.
84. Search unusual network activity.
85. Validate results.

86. Record hunt outcome and next steps.

Microsoft / Lab Tools

Defender Advanced Hunting, Sentinel, KQL

Expected Learning Outcome

Students can perform structured threat hunting.

Module 16: MITRE ATT&CK Mapping & Detection Thinking — 2 Hours

Theory Concepts

- Tactics and techniques
- Technique evidence
- Mapping investigation findings
- Detection coverage
- Reporting attacker behavior

Hands-On Lab Objective

Map simulated Microsoft security findings to ATT&CK techniques.

Practical Lab Activities

87. Map PowerShell behavior.
88. Map credential-access behavior.
89. Map discovery behavior.
90. Map lateral movement behavior.
91. Add ATT&CK references to the final report.

Microsoft / Lab Tools

MITRE ATT&CK, Defender, Sentinel

Expected Learning Outcome

Students can communicate attacker behavior consistently.

Module 17: Incident Response & Automated Response — 4 Hours

Theory Concepts

- Containment and eradication
- Endpoint isolation
- Account response
- Email remediation
- Automation rules
- Playbooks/Logic Apps concepts
- Human approval

Hands-On Lab Objective

Design and execute a safe response workflow in the lab.

Practical Lab Activities

92. Identify response options.
93. Practice endpoint isolation where supported.
94. Review account response choices.

95. Create an automation design.
96. Define approval checkpoints.
97. Document rollback considerations.

Microsoft / Lab Tools

Microsoft Defender XDR, Microsoft Sentinel, Logic Apps concepts

Expected Learning Outcome

Students understand safe automated and manual response.

Module 18: Security Automation & AI Assistance — 2 Hours

Theory Concepts

- SOC automation principles
- Python and APIs
- Structured incident data
- AI-assisted summarization
- Human verification
- Automation safety

Hands-On Lab Objective

Build a small local analyst assistant that summarizes a synthetic Microsoft security alert.

Practical Lab Activities

98. Export/create sample alert JSON.
99. Parse fields with Python.
100. Generate a structured summary.
101. Optionally use Ollama/local AI.
102. Verify every generated conclusion against source data.

Microsoft / Lab Tools

Python, Ollama, JSON, Microsoft security sample data

Expected Learning Outcome

Students can safely use automation/AI to reduce repetitive analyst work.

Module 19: End-to-End Microsoft SOC Challenge — 4 Hours

Theory Concepts

- Multi-stage incident correlation
- Identity + endpoint + email + SIEM
- Triage
- Threat hunting
- Containment
- Reporting

Hands-On Lab Objective

Investigate a simulated multi-stage attack using the Microsoft security stack.

Practical Lab Activities

103. Start from an initial alert.

104. Correlate identity and endpoint evidence.
105. Investigate email or network evidence.
106. Use KQL hunting.
107. Determine scope.
108. Recommend containment.
109. Map behaviors.
110. Prepare an executive summary.

Microsoft / Lab Tools

Defender XDR, Defender for Endpoint, Sentinel, KQL, MITRE ATT&CK

Expected Learning Outcome

Students demonstrate integrated Microsoft SOC investigation skills.

Module 20: Final Capstone: Microsoft Security Operations Analyst — 3 Hours

Theory Concepts

- Security operations architecture
- Detection
- Investigation
- Threat hunting
- Incident response
- Reporting
- Operational handover

Hands-On Lab Objective

Complete a portfolio-style Microsoft SOC investigation from alert through final report.

Practical Lab Activities

111. Configure a small lab.
112. Create or ingest telemetry.
113. Build at least one detection.
114. Investigate one confirmed incident.
115. Perform a hunting query.
116. Document evidence.
117. Recommend remediation.
118. Deliver a final SOC handover.

Microsoft / Lab Tools

Microsoft Defender, Microsoft Sentinel, KQL, Microsoft Entra ID, MITRE ATT&CK

Expected Learning Outcome

Students complete an end-to-end Microsoft security operations workflow.

Final Assessment Plan

Component	Weight	Evidence
Microsoft Security Knowledge Checks	15%	Quizzes and short scenario questions
Hands-On Labs	25%	Configuration, queries, investigation evidence
KQL / Hunting Assessment	15%	Independent KQL investigation
Incident Investigation	20%	Defender + Sentinel investigation report
Final Microsoft SOC Capstone	25%	End-to-end detection, investigation, response and handover

Student Deliverables

- Microsoft security architecture diagram
- Entra ID identity investigation worksheet
- Defender incident investigation report
- Defender for Endpoint device investigation
- Phishing investigation report
- Sentinel data-onboarding worksheet
- KQL query notebook
- Sentinel detection rule
- Sentinel incident investigation report
- Advanced Hunting exercise
- MITRE ATT&CK mapping
- Threat-hunting report
- Incident-response plan
- Final SOC incident report
- Final SOC handover presentation

Beginner-Friendly Teaching Method

- Explain the security scenario before introducing the Microsoft feature.
- Demonstrate every portal workflow once, then let students repeat it.
- Teach KQL from simple filtering before moving to joins and advanced hunting.
- Use one simulated incident across multiple modules so students learn correlation.
- Keep trial data ingestion small and monitor Azure usage.
- Use a dedicated trial/lab tenant; do not experiment with production security policies.
- Require evidence for every practical investigation: query, screenshot, timeline or report.
- Teach response actions conceptually first; use destructive/isolation actions only in controlled labs.

Trial & Cost-Control Rules

- Create a dedicated training tenant whenever possible.
- Use synthetic users and test devices.
- Keep Sentinel ingestion within the documented trial allowance.
- Monitor Azure Cost Management throughout the course.

- Remove or stop chargeable Azure resources after each batch.
- Do not connect student labs to production Microsoft 365 environments.
- Verify trial eligibility and feature availability before promising a specific lab.
- Where a trial feature is unavailable, use Microsoft Learn simulated/lab material or local sample data rather than requiring paid licensing.

Important Current Microsoft Platform Note

Microsoft Sentinel is now available in the Microsoft Defender portal, and Microsoft states that Sentinel does not require Defender XDR or an E5 license when used there. Microsoft also states that after March 31, 2027, Sentinel will no longer be supported in the Azure portal and will be available only in the Defender portal. This course therefore teaches the Defender-portal experience alongside the Azure concepts needed for the current trial environment. □cite□turn0search17□turn0search19□

Microsoft's Defender documentation recommends trial environments for hands-on learning, and its Defender XDR lab exercises explicitly support trial licensing for the training environment. □cite□turn0search1□turn0search2□

Suggested Course Name & Tagline

Course Name: Microsoft Security Operations & Defender Sentinel Practitioner

Tagline: “LEARN MICROSOFT SECURITY → INVESTIGATE → HUNT → RESPOND”